

## Lesson 11 · Profit optimisation and feasible decisions

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.  
Work at your own pace; partial completion is welcome.

### Q1 · Rank 1 / Foundation

A firm has demand  $P = aQ + b$  ( $a < 0$ ,  $b > 0$ ), fixed costs  $c$  and variable costs  $d$  per unit. Write down a general expression for total revenue.

☐ A  $aQ + b$

☐ B  $aQ^2 + bQ$

☐ C  $aQ^2 + b$

☐ D  $c + dQ$

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### Q2 · Rank 1 / Foundation

A firm has demand  $P = aQ + b$  ( $a < 0$ ,  $b > 0$ ), fixed costs  $c$  and variable costs  $d$  per unit. Write down a general expression for total cost.

☐ A  $c + d$

☐ B  $cQ + d$

☐ C  $c + dQ$

☐ D  $dQ$

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### Q3 · Rank 1 / Foundation

A firm has demand  $P = aQ + b$  ( $a < 0$ ,  $b > 0$ ), fixed costs  $c$  and variable costs  $d$  per unit. Deduce the marginal cost.

☐ A  $c$ ☐ B  $c + d$ ☐ C  $dQ$ ☐ D  $d$ 

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### Q4 · Rank 2 / Core

A firm has demand  $P = aQ + b$  ( $a < 0$ ,  $b > 0$ ), fixed costs  $c$  and variable costs  $d$  per unit. Deduce the marginal revenue.

☐ A  $2aQ + b$ ☐ B  $aQ + b$ ☐ C  $2a + b$ ☐ D  $aQ^2 + bQ$ 

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### Q5 · Rank 2 / Core

A firm has demand  $P = aQ + b$  ( $a < 0$ ,  $b > 0$ ), fixed costs  $c$  and variable costs  $d$  per unit. Assume also  $b > d \geq 0$  and  $c \geq 0$ . Find the non-negative output which maximises profit.

☐ A  $\frac{b-d}{2a}$

☐ B  $\frac{d-b}{2a}$

☐ C  $\frac{d-b}{a}$

☐ D  $\frac{-b}{2a}$

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### Q6 · Rank 2 / Core

A firm has  $TC = Q^2 + 50Q + 10$  and demand  $P = 200 - 4Q$ . Find the level of output needed to maximise profit.

☐ A 25

☐ B 30

☐ C 15

☐ D 50

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### Q7 · Rank 3 / Application

A firm has demand  $P = 60 - 0.5Q$ , fixed costs 10 and variable costs  $Q + 3$  per unit. Find the maximum profit.

☐ A 541.5☐ B 531.5☐ C 1083☐ D 19

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### Q8 · Rank 3 / Application

The average cost per person of hiring a tour guide for a week's river cruise is  $AC = 3Q^2 - 192Q + 3500$ , where  $0 < Q \leq 30$ . Find the minimum average cost for the trip.

☐ A 428☐ B 3500☐ C 32☐ D 440

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### Q9 · Rank 3 / Application

A firm has production  $Q = 3\sqrt{L}$ , sells everything at USD 50 per unit, and pays USD 10 per unit of labour. Ignore other costs. Find the value of  $L$  which maximises profit, treating labour as continuous.

☐ A 56.25☐ B 7.5☐ C 225☐ D 0

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### Q10 · Rank 4 / Challenge · OPTIONAL

A firm has  $TC = Q^2 + 50Q + 10$  and  $P = 200 - 4Q$ . A tax of  $t$  dollars per good is added to its costs. For  $0 \leq t < 150$ , the firm continues to maximise profit. By how much does the price increase?

☐ A  $t$ ☐ B  $t/5$ ☐ C  $2t/5$ ☐ D  $3t/5$ 

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